

Lecture 10 — Pricing Barrier Options

Study notes: the pricing problem, analytical GBM results, and COS for discrete barriers

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Abstract

Barrier options pay only if the underlying does (knock-in) or does not (knock-out) breach a level over $[0, T]$. These notes cover the three routes in the lecture: (1) the *pricing problem* — the eight barrier types, in-out parity, and the two equivalent formulations (a first-passage expectation and a localized Feynman–Kac PDE with absorbing boundaries); (2) *analytical results under GBM* — the reflection principle (derived), the density of the running maximum and the joint density of (W_T, Ξ_0^T) , the closed-form up-and-out call, and two PDE routes (method of images and a sine-series heat-equation solution); (3) *the COS method for discretely-monitored barriers* — the backward recursion over monitoring dates, COS applied at each step, the backward induction for the cosine coefficients V_k , the Hankel + Toeplitz structure of the coefficient update and its FFT acceleration to $O((M-1)N \log_2 N)$, the terminal conditions, and the truncation range. The key conceptual point: for a *discrete* barrier COS becomes a backward recursion and, unlike European COS, *does* use the FFT, because each step is a structured matrix–vector product. Prereq: the European COS machinery (notes/Lecture06-notes.pdf); $\sum'$ halves the $k = 0$ term.

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1 The pricing problem

1.1 Definitions: eight barrier types

Fix a barrier level B (or H) and maturity T . A barrier option is a vanilla call/put whose existence is gated by whether the asset path touches B during monitoring:

- **up-and-out call**: pays $\max(S_T - K, 0)$ *unless* the asset crosses $B > S_0$ at some time in $[0, T]$, in which case it pays 0 (it is “knocked out”);
- **up-and-in call**: pays 0 *unless* the asset crosses $B > S_0$, in which case it becomes a vanilla call (“knocked in”).

Replacing up $\leftrightarrow$ down and call $\leftrightarrow$ put gives $2 \times 2 \times 2 = 8$ types (up/down $\times$ in/out $\times$ call/put). Each admits an analytical formula under GBM.

1.2 In–out parity

Owning the knock-in *and* the knock-out with the same (K, B, T) reproduces the vanilla, because exactly one of the two is alive at maturity:

$$C_{\text{up-in}}(t) + C_{\text{up-out}}(t) = e^{-r(T-t)} \mathbb{E}[(S_T - K)^+ | \mathcal{F}_t], \quad (1)$$

and likewise for down/in–out and for puts. *Practical use*: price the easier one (often the knock-out) plus the vanilla, then back out the other by subtraction. It also gives a free correctness check on any numerical barrier price.

1.3 Two equivalent formulations

Let $X_t = \ln S_t$ (or a log-return), τ_B the first time the barrier is breached, and $\alpha = +1$ for a call, -1 for a put. The **expectation (martingale) route** writes the (e.g. up-and-out) price as a discounted expectation with a survival indicator:

$$V(x, t) = e^{-r(T-t)} \mathbb{E}_{X_t=x} \left[(\alpha (e^{X_T} - K))^+ \cdot \mathbf{1}_{\{\tau_B > T\}} \right]. \quad (2)$$

The **PDE route** (localized Feynman–Kac): on $x \in [a, b]$ the same V is the unique bounded solution of

$$\frac{\partial V}{\partial t} + \left(r - \frac{1}{2}\sigma^2\right) \frac{\partial V}{\partial x} + \frac{1}{2}\sigma^2 \frac{\partial^2 V}{\partial x^2} - rV = 0, \quad (3)$$

with terminal data $V(x, T) = (\alpha(e^x - K))^+$ and *absorbing (Dirichlet) boundary conditions* $V(a, t) = V(b, t) = 0$ at the knock-out level(s). The barrier enters the PDE not through the equation but through the homogeneous boundary condition: hitting the barrier sets the value to zero. The two routes are the same object; (2) is convenient for Monte Carlo and for the COS method, (3) for finite differences and for the analytic GBM solutions of §2.

2 Analytical results under GBM

Under GBM the asset is $S_t = S_0 \exp(\sigma W_t + (r - \frac{1}{2}\sigma^2)t)$, so the barrier event depends on the running extreme of a (drifted) Brownian motion. The closed forms all rest on the distribution of that extreme.

2.1 The running maximum and the reflection principle

Let W be standard BM, $W_0 = 0$, and $\Xi_0^T = \max_{t \in [0, T]} W_t \geq 0$. For a level $a > 0$ let $\tau_a = \inf\{t \geq 0 : W_t = a\}$ be the first hitting time. The two events $\{\Xi_0^T \geq a\}$ and $\{\tau_a \leq T\}$ are identical.

Reflection. By the strong Markov property at τ_a and the symmetry of BM, a path that has already touched a is equally likely to finish above or below a : $\mathbb{P}(W_T \geq a \mid \tau_a \leq T) = \mathbb{P}(W_T \leq a \mid \tau_a \leq T) = \frac{1}{2}$. Hence

$$\begin{aligned}\mathbb{P}(\tau_a \leq T) &= \mathbb{P}(\tau_a \leq T, W_T > a) + \mathbb{P}(\tau_a \leq T, W_T \leq a) \\ &= 2\mathbb{P}(\tau_a \leq T, W_T \geq a) = 2\mathbb{P}(W_T \geq a),\end{aligned}\quad (4)$$

where the last step uses $\{W_T \geq a\} \subseteq \{\tau_a \leq T, W_T \geq a\} \subseteq \{W_T \geq a\}$ (to reach a level above a at T you must have hit a earlier). Therefore the maximum has the law of $|W_T|$:

$$\mathbb{P}(\Xi_0^T \leq a) = 1 - 2\mathbb{P}(W_T \geq a) = \mathbb{P}(|W_T| \leq a) = \frac{2}{\sqrt{2\pi T}} \int_0^a e^{-x^2/2T} dx, \quad (5)$$

with density

$$\boxed{p_{\Xi_0^T}(a) = \sqrt{\frac{2}{\pi T}} e^{-a^2/2T} \mathbf{1}_{[0,\infty)}(a).} \quad (6)$$

A consequence: $\mathbb{P}(W_t \leq 0 \ \forall t \in [0, \varepsilon]) = \mathbb{P}(\Xi_0^\varepsilon = 0) = \int_0^0 p = 0$, i.e. BM is immediately positive somewhere with probability one.

2.2 The joint density of (Ξ_0^T, W_T)

Reflect the terminal point across the level. For $b \leq a$, a path with $\Xi_0^T \geq a$ ending below b is in bijection (reflect after τ_a) with a path ending above $2a - b$:

$$\mathbb{P}(\Xi_0^T \geq a, W_T < b) = \mathbb{P}(\Xi_0^T \geq a, W_T > 2a - b) = \mathbb{P}(W_T > 2a - b), \quad (7)$$

the last equality because $2a - b \geq a$ forces the maximum to exceed a . So $\mathbb{P}(\Xi_0^T > a, W_T \leq b) = \mathbb{P}(W_T \geq 2a - b) = \frac{1}{\sqrt{2\pi T}} \int_{2a-b}^\infty e^{-x^2/2T} dx$, and differentiating,

$$p_{\Xi_0^T, W_T}(a, b) = -\frac{\partial^2}{\partial a \partial b} \mathbb{P}(\Xi_0^T > a, W_T \leq b) = \sqrt{\frac{2}{\pi}} \frac{2a - b}{T^{3/2}} e^{-(2a-b)^2/2T} \mathbf{1}_{\{a \geq \max(b, 0)\}}. \quad (8)$$

For *drifted* BM one obtains the law by a Girsanov change of measure (the drift becomes a Radon–Nikodym exponential weight $e^{\theta W_T - \frac{1}{2}\theta^2 T}$). Equation (8) is exactly the $p(\xi, y)$ that appears in the expectation route below.

2.3 Closed-form barrier prices

Any payoff $\phi(S_T, \max_\tau S_\tau)$ integrates against (8):

$$V(x, t) = e^{-r(T-t)} \int_{\mathbb{R}} \int_{\mathbb{R}} \phi(S_t e^{\sigma y + (r - \frac{1}{2}\sigma^2)(T-t)}, S_0 e^{\sigma \xi + (r - \frac{1}{2}\sigma^2)(T-t)}) p(\xi, y) d\xi dy. \quad (9)$$

For the **up-and-out call** this evaluates to the standard form

$$\begin{aligned}C_{\text{up-out}} &= S_t \left[N(d_1) - N(e_1) - (B/S_t)^{1+2r/\sigma^2} (N(f_2) - N(g_2)) \right] \\ &\quad - e^{-rT} K \left[N(d_2) - N(e_2) - (B/S_t)^{-1+2r/\sigma^2} (N(f_1) - N(g_1)) \right],\end{aligned}\quad (10)$$

with $d_{1,2}$ the usual Black–Scholes arguments and the barrier-shifted arguments

$$\begin{aligned}e_{1,2} &= \frac{1}{\sigma\sqrt{T-t}} \left[\ln \frac{S_t}{B} + (r \pm \frac{1}{2}\sigma^2)(T-t) \right], \quad f_{2,1} = \frac{1}{\sigma\sqrt{T-t}} \left[\ln \frac{S_t}{B} - (r \pm \frac{1}{2}\sigma^2)(T-t) \right], \\ g_{2,1} &= \frac{1}{\sigma\sqrt{T-t}} \left[\ln \frac{S_t K}{B^2} - (r \pm \frac{1}{2}\sigma^2)(T-t) \right].\end{aligned}$$

The $N(e), N(f), N(g)$ terms are precisely the reflected/image contributions that subtract the barrier-breaching paths; the $(B/S_t)^{\pm 1+2r/\sigma^2}$ powers are the GBM analogue of the image weight derived next.

2.4 PDE route 1 — method of images

On the BS operator $\mathcal{L}V = V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV$, one checks directly that if $\mathcal{L}V = 0$ then so does the *image*

$$U(S, t) = S^{2\alpha} V\left(\frac{B^2}{S}, t\right), \quad \alpha = \frac{1}{2}\left(1 - \frac{2r}{\sigma^2}\right). \quad (11)$$

This is the BS analogue of reflecting a heat-equation solution across a wall to enforce a homogeneous boundary (method of images). Subtracting the image with the right weight cancels the value on the barrier $S = B$, yielding e.g. the **down-and-out call**

$$C_{\text{down-out}}(S_t, K) = C_{\text{vanilla}}(S_t, K) - \left(\frac{S_t}{B}\right)^{2\alpha} C_{\text{vanilla}}\left(\frac{B^2}{S_t}, K\right). \quad (12)$$

The first term is the vanilla; the second removes exactly the paths that would have crossed B , evaluated at the mirror spot B^2/S_t with the image weight $(S_t/B)^{2\alpha}$.

2.5 PDE route 2 — sine-series solution of the heat equation

This route is the spiritual precursor of the COS barrier method and worth working through. The localized BS PDE (3) on $x \in [a, b]$ with $V(a, t) = V(b, t) = 0$ reduces to the heat equation. Substitute

$$\tau = \frac{1}{2}\sigma^2(T-t), \quad \kappa = \frac{2r}{\sigma^2}, \quad V(x, \tau) = e^{\alpha x + \beta \tau} U(x, \tau), \quad \alpha = -\frac{1}{2}(\kappa - 1), \quad \beta = -\alpha^2 - \kappa, \quad z = x - a, \quad (13)$$

giving, for $z \in [0, b - a]$,

$$\frac{\partial U}{\partial \tau} = \frac{\partial^2 U}{\partial z^2}, \quad U(0, \tau) = U(b - a, \tau) = 0, \quad U(z, 0) = g(z) = e^{-\alpha(z+a)} K(e^{z+a} - 1)^+. \quad (14)$$

The homogeneous Dirichlet conditions select the *sine* basis. Expand the initial datum,

$$g(z) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi z}{b-a}\right), \quad b_n = \frac{2}{b-a} \int_0^{b-a} g(z) \sin\left(\frac{n\pi z}{b-a}\right) dz, \quad (15)$$

and propagate each mode by its heat eigenvalue $e^{-(n\pi/(b-a))^2 \tau}$:

$$U(z, \tau) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi z}{b-a}\right) e^{-(n\pi/(b-a))^2 \tau}. \quad (16)$$

Undoing the substitutions gives the barrier price as a sine series:

$$V(x, t) = e^{-r(T-t)} \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi(x-a)}{b-a}\right) e^{\alpha x} e^{-\frac{1}{2}\sigma^2(T-t) \left[(n\pi/(b-a))^2 + \alpha^2 \right]}. \quad (17)$$

Note the structural parallel with COS: a finite-interval Fourier basis, coefficients from the payoff (b_n), and an exponential “propagator”. The difference is that here the propagator is the explicit GBM/heat semigroup, whereas COS uses the characteristic function and so generalises to any model with a known φ .

3 COS for discretely-monitored barriers

3.1 Why discrete monitoring, and why a backward recursion

In practice barriers are checked on a finite grid $t_1 < \dots < t_M = T$. Write $x_m = \ln(S(t_m)/K)$ and $h = \ln(H/K)$. Knock-out survival is a *product* of indicators,

$$\mathbf{1}_{\{\tau > T\}} = \prod_{m=1}^M \mathbf{1}_{\{x_m < h\}}, \quad (18)$$

and (up-and-out, rebate R) the price is

$$V(X_0, t_0) = \mathbb{E}^{\mathbb{Q}} \left[V(X_M, t_M) \prod_{m=1}^M \mathbf{1}_{\{x_m < h\}} + e^{-r\tau} R \mathbf{1}_{\{\tau \leq T\}} \mid X_0 \right]. \quad (19)$$

By the Markov property the joint density factorises into one-step transitions, $f(x_M, \dots, x_1 \mid x_0) = f(x_M \mid x_{M-1}) \cdots f(x_1 \mid x_0)$, so the M -dimensional integral (19) nests:

$$\int_{\mathbb{R}} f(x_1 \mid x_0) \mathbf{1}_{\{x_1 < h\}} \cdots \int_{\mathbb{R}} f(x_{M-1} \mid x_{M-2}) \mathbf{1}_{\{x_{M-1} < h\}} \int_{\mathbb{R}} V(x_M) f(x_M \mid x_{M-1}) \mathbf{1}_{\{x_M < h\}} dx_M \cdots dx_1. \quad (20)$$

This is precisely a **backward recursion**. Define the continuation value and the (knock-out) option value at each monitoring date, for $m = M, M-1, \dots, 2$:

$$c(x, t_{m-1}) = e^{-r\Delta t} \int_{\mathbb{R}} v(y, t_m) f(y \mid x) dy, \quad v(x, t_{m-1}) = \begin{cases} e^{-r(T-t_{m-1})} R, & x \geq h \text{ (knocked out)}, \\ c(x, t_{m-1}), & x < h \text{ (still alive)}, \end{cases} \quad (21)$$

terminating with $v(x, t_0) = e^{-r\Delta t} \int v(y, t_1) f(y \mid x) dy$. Each step is one continuation integral of exactly the European COS form — the lecture's whole construction is to evaluate (21) by COS, fast, $M-1$ times.

3.2 The COS step

Apply the European COS density expansion (Lecture 6) to $f(y \mid x)$ inside (21). With

$$F_k(x) = \operatorname{Re} \left[\varphi \left(\frac{k\pi}{b-a}; x \right) e^{-ik\pi a/(b-a)} \right], \quad V_k(t_m) = \frac{2}{b-a} \int_a^b v(y, t_m) \cos \left(\frac{k\pi(y-a)}{b-a} \right) dy,$$

the continuation value is the familiar finite sum

$$c(x, t_{m-1}) \approx \hat{c}(x, t_{m-1}) = e^{-r\Delta t} \sum_{k=0}^{N-1'} F_k(x) V_k(t_m). \quad (22)$$

Here $\varphi(\omega; x)$ is the *conditional* characteristic function of y given x ; for a Lévy process the increment is independent of x , so $\varphi(\omega; x) = \varphi_{\text{Lévy}}(\omega) e^{i\omega x}$, which is what makes the x -dependence cheap. The new ingredient versus Lecture 6: the “payoff” $v(y, t_m)$ is itself produced by the previous COS step, so its coefficients V_k must be propagated.

3.3 Backward induction for the coefficients V_k

Because v is defined piecewise at h (21), its cosine coefficient splits at the barrier:

$$\hat{V}_k(t_m) = \underbrace{\frac{2}{b-a} \int_a^h \hat{c}(y, t_m) \cos \left(\frac{k\pi(y-a)}{b-a} \right) dy}_{\hat{C}_k(a, h, t_m)} + \underbrace{\frac{2e^{-r(T-t_{m-1})} R}{b-a} \int_h^b \cos \left(\frac{k\pi(y-a)}{b-a} \right) dy}_{e^{-r(T-t_{m-1})} \frac{2R}{b-a} \Psi_k(h, b)}. \quad (23)$$

The cosine integral $\Psi_k(h, b) = \int_h^b \cos \left(\frac{k\pi(y-a)}{b-a} \right) dy$ is elementary (the ψ_k of Lecture 6). The first piece, inserting the COS continuation value (22), becomes a sum over the previous coefficients:

$$\hat{C}_k(x_1, x_2, t_m) = e^{-r\Delta t} \operatorname{Re} \left[\sum_{j=0}^{N-1'} \varphi_{\text{Lévy}} \left(\frac{j\pi}{b-a} \right) V_j(t_{m+1}) M_{k,j}(x_1, x_2) \right], \quad (24)$$

where the coupling integral is analytic:

$$M_{k,j}(x_1, x_2) = \frac{2}{b-a} \int_{x_1}^{x_2} e^{ij\pi(x-a)/(b-a)} \cos\left(\frac{k\pi(x-a)}{b-a}\right) dx. \quad (25)$$

Writing $\cos = \frac{1}{2}(e^i + e^{-i})$ splits $M_{k,j} = \frac{1}{2}(M_{k,j}^c + M_{k,j}^s)$ into a piece depending on $j+k$ and one on $j-k$:

$$M_{k,j}^c = \frac{\exp(i(j+k)\frac{(x_2-a)\pi}{b-a}) - \exp(i(j+k)\frac{(x_1-a)\pi}{b-a})}{(j+k)}, \quad M_{k,j}^s = \frac{\exp(i(j-k)\frac{(x_2-a)\pi}{b-a}) - \exp(i(j-k)\frac{(x_1-a)\pi}{b-a})}{(j-k)}, \quad (26)$$

(with the removable cases $j+k=0$ and $j=k$ giving the linear term $(x_2-x_1)\pi i/(b-a)$). The summary recursion is then

$$\hat{V}_k(t_m) = \hat{C}_k(a, h, t_m) + e^{-r(T-t_{m-1})} \frac{2R \Psi_k(h, b)}{b-a}. \quad (27)$$

Terminal condition. At t_M the coefficients are those of the actual payoff $v(y, t_M)$; with G_k the cosine coefficients of the vanilla payoff $K(\alpha(e^y - 1))^+$ (the $\chi_k - \psi_k$ combinations of Lecture 6) one has, e.g. for $h \geq 0$,

$$V_k(t_M) = \begin{cases} G_k(0, h) + \frac{2R \Psi_k(h, b)}{b-a}, & \text{call,} \\ G_k(a, 0), & \text{put,} \end{cases}$$

and the analogous split for $h < 0$.

3.4 Hankel + Toeplitz structure and the FFT

For a fixed pair (x_1, x_2) the matrix $M_{k,j}^c$ depends only on $j+k$ (a **Hankel** matrix) and $M_{k,j}^s$ only on $j-k$ (a **Toeplitz** matrix). Define $u_j = \varphi(\frac{j\pi}{b-a})V_j(t_{m+1})$ (with u_0 halved). The required product is

$$\hat{C}(x_1, x_2, t_m) = \frac{e^{-r\Delta t}}{\pi} \operatorname{Im} \{ (\mathcal{M}_c + \mathcal{M}_s) u \}. \quad (28)$$

A Toeplitz (or Hankel) matrix times a vector is a *discrete convolution*, which by circulant embedding is computed with three FFTs:

$$x \otimes y = \mathcal{D}^{-1} \{ \mathcal{D}(x) \cdot \mathcal{D}(y) \}. \quad (29)$$

This drops each step from the naive $O(N^2)$ to $O(N \log_2 N)$. Over the $M-1$ monitoring dates the **overall cost is**

$$\boxed{O((M-1) N \log_2 N)}. \quad (30)$$

Remark 1 (The FFT returns — a key contrast with Lecture 6). *European COS is a single $O(N)$ cosine sum and uses no FFT. The discrete-barrier method needs the coefficients $\{V_k(t_m)\}$ at every step, and propagating them is a structured (Toeplitz/Hankel) matrix-vector product across the whole coefficient vector — that is the convolution the FFT accelerates. So the FFT, dismissed for the European problem, is exactly what makes the barrier recursion tractable.*

3.5 The algorithm

1. Choose $[a, b]$ (§3.6) and set the terminal coefficients $\hat{V}_k(t_M) = V_k(t_M)$.
2. For $m = M, M-1, \dots, 2$:

- (a) form $u_j = \varphi(\frac{j\pi}{b-a})\hat{V}_j(t_m)$;
- (b) compute $\hat{C}_k(a, h, t_m)$ via the Hankel/Toeplitz-vector products by FFT;
- (c) assemble $\hat{V}_k(t_{m-1}) = \hat{C}_k(a, h, t_{m-1}) + e^{-r(T-t_{m-1})} \frac{2R\Psi_k(h,b)}{b-a}$.

3. Final price at t_0 : $\hat{v}(x, t_0) = e^{-r\Delta t} \sum_{k=0}^{N-1} F_k(x) \hat{V}_k(t_1)$.

“In” barriers follow from in-out parity (1) (price the out-version, subtract from vanilla), which is cheaper and numerically more stable than handling the in-condition directly.

3.6 The truncation range

Same cumulant rule as Lecture 6, but now centred with the log-moneyness $x_0 = \ln(S_0/K)$:

$$[a, b] = \left[(c_1 + x_0) - L\sqrt{c_2 + \sqrt{c_4}}, (c_1 + x_0) + L\sqrt{c_2 + \sqrt{c_4}} \right], \quad (31)$$

with $c_1, \dots, c_4$ the cumulants of the (one-step) log-increment and L fixed by tolerance. The trade-off is sharp here because the range is fixed once and reused for all M steps: too small $\Rightarrow$ integration-range truncation error; too large $\Rightarrow$ large N for the same accuracy. The same short-maturity caution as in Lecture 6 applies to each step ($c_2 \propto \Delta t$ small), and additionally the range must be wide enough that the barrier h sits well inside $[a, b]$ — otherwise the split at h in (23) is meaningless. Empirically (CGMY tests, slide 35) the method reaches $\sim 10^{-8}$ error at $N = 2^7$ in a few milliseconds per option, with the error ratio confirming high-order convergence in N .

3.7 Continuous monitoring and Heston

- **Continuous monitoring** is the $M \rightarrow \infty$ (or $\Delta t \rightarrow 0$) limit; in practice the discrete price converges to it, and a Richardson-type extrapolation in Δt (with the known $O(\sqrt{\Delta t})$ discrete-vs-continuous barrier bias) recovers the continuous price. The clean GBM continuous case is §2.
- **Heston.** The same backward recursion applies, but the state is now (log-stock, variance), so the continuation integral is two-dimensional: either 1-D COS in the log-stock with numerical integration over variance, or a full 2-D COS recovering the joint density. The characteristic function is Heston’s (Lecture 7); the Feller condition $2\kappa\bar{v} \geq \eta^2$ keeps it well-behaved.

4 One-paragraph reconstruction (say this out loud)

A barrier option pays the vanilla payoff times a survival indicator. Continuously and under GBM, the survival event is governed by the running maximum of Brownian motion, whose law I get from the reflection principle ($p_{\Xi}(a) = \sqrt{2/\pi T} e^{-a^2/2T}$, joint density $\propto (2a-b)e^{-(2a-b)^2/2T}$), giving closed-form barrier prices; equivalently the localized BS PDE with absorbing boundaries at the barrier, solved by images ($U = S^{2\alpha} V(B^2/S, t)$) or by a sine series of the heat equation. For a *discretely*-monitored barrier under a general model with known φ , I write the price as a backward recursion over monitoring dates: at each date the continuation value is $\hat{c}(x, t_{m-1}) = e^{-r\Delta t} \sum_k F_k(x) V_k(t_m)$ (European COS), then I knock out by overwriting $v = R$ above h , which splits the coefficient V_k at the barrier into a payoff piece $\Psi_k(h, b)$ and a continuation piece $\hat{C}_k$ built from the previous V_j through the analytic coupling $M_{k,j}$. That coupling is Hankel + Toeplitz, so each step is a convolution done by FFT in $O(N \log N)$, and the whole price costs $O((M-1)N \log_2 N)$. Truncation range is the cumulant box $(c_1 + x_0) \pm L\sqrt{c_2 + \sqrt{c_4}}$, wide enough to contain the barrier.